Comp 3106 A Project Proposal

Ho Nam Leung 101197127

Title: Developing an AI to play Big Two

Introduction

Big Two is a well-known card game of Cantonese origin. In the game a typical deck of playing cards is used, and 4 players will take turns playing their card or passing the turn, and the goal is to be the first to play off all of one’s hand. In the game there is a rule set on what cards can be played in what situations, and in certain situations multiple cards can be / must be played. In this project, I aim to develop an AI that will learn how to play the card game, from simple straight forward playstyles to complicated strategies that mimics how an actual person will play the game. The detailed rules for the game will be included at the end of the document.

Proposed objectives

In this project, I aim to create an AI that will be able to play the card game Big Two. There will be 2 main objectives for the project.

First, the AI will have to be able to complete one round of the game. This does not require complicated strategies or playstyles, it just have to be able to simply move towards the end of the round and should know how to play different cards or pass the turn in appropriate situations. The AI is expected to have a very low winning probability in this case.

Second, the AI will have to learn strategies in the game that will lead to a higher rate of winning the game. It should not just play any cards when possible, but save cards to form bigger sets. The AI should have a higher winning probability compared to the first objective.

Proposed Methods

For the first objective, I should be able to use a model-based AI to complete the task. I will have to create a model where it finds any suitable cards to play in a certain situation, then play that card.

For the second objective, I will be using reinforcement learning with rewards and punishments for different situations to help the AI learn what strategies will lead to better winning percentages. Possible rewards might be winning the game, discarding a card, discarding multiple cards, make opponents pass etc. Possible punishments might be losing the game, passing a turn, breaking up a big set of cards etc.

Dataset to be used

I will be creating a Big Two game using Godot engine with an API that allows one or multiple AIs to connect to and control the game. The game should also allow human input. I will be combining both automatic and manual training for the AI (competing against humans or against other AIs).

Proposed validation/analysis strategy

For the validation stage, the AI will only be competing with humans, and human players will try their best to win. The AI should have a decent win percentage, or at least pose a challenge for the human players by trying to stop them from winning.

Description of the novelty of the project

Despite Big Two being well known in Asia, I don't think I have heard about it a lot in western countries. Therefore, I think it would be interesting to create an AI that will be able to play this game. Furthermore, having a simple AI that is able to play the game is not hard (first objective), but I wish to create an AI that will have more complicated strategies and pose more of a challenge when playing against other players.

Weekly schedule

Nov 3 – 16 (2 weeks): Create a game of Big Two using Godot with an API that allows the AI to connect to and control the game. The game should also accept human input.

Nov 17 – 23 (1 week): Create a simple AI that is able to complete the game (first objective)

Nov 24 – 30 (1 week): Create reinforcement learning for the AI and train the AI to get better at playing (second objective)

Dec 1 – 6 (1 week): Complete the project report

Availability for project demonstration

Preference: online

Availability: 1/12 12:00pm – 7:00pm, 2/12 4:00pm – 5:30pm, 3/12 12:00pm – 7:00pm

GPU resources from school

I do not require GPU resources from school

Detailed rules for Big Two

The entire deck is dealt. The cards are dealt out among the players, retaining an equal number of cards for each player. The Joker cards are not used.

At the beginning of each game, the player with the diamond 3 starts by playing it singly or as part of a combination. Play proceeds; each player must play a higher card or combination than the one before, with the same number of cards. Players may also pass, declaring that they do not want to play, or do not hold the necessary cards to make a play possible.

When all but one of the players have passed in succession, the turn is over. A new turn is started with all players, initiated by the last player to play.

The game ends when one player runs out of cards.

Combinations:

Singles, Pairs, Triples, Five-cards

Five-cards (ascending order): Straight, Flush, Full house, Four of a kind, Straight Flush

(from Wikipedia: https://en.wikipedia.org/wiki/Big_two)